

Data governance project

Introduction:

In this project, you will work in **teams of 3 or 4 students** who are required to work together on their respective datasets that you choose to work on within the project

Project phases:

Please revise the following schedule:

Phase	Deadline
Phase 1 (5 Points)	Week 10 14/4
Phase 2 (5 Points)	Week 12 28/4
Discussion (5 Points)	Week 13 5/5

Phase 1: Dataset

- **First**, you need to find a dataset that contains **textual data, passwords/phone numbers, and numerical data**, and be aware that your data needs to be **approved by the assigned TA**.
The datasets must consist of **over 10,000 records** and clearly exhibit issues requiring resolution.
Each team must have **a unique dataset**.

It is **NOT ALLOWED** to use **AI-generated data**. The data needs to be real and has a degree of correlation.

- **Second**, you will need to **apply one of the data profiling techniques** to your chosen data and use the data you get from it to **make your own copy of the [profiling report](#)** of phase 1.
- **Each team needs to fill in this [Google Sheet](#) with team members' info, the dataset's name, the dataset's link, the dataset's description, and an identification of the dataset's issues.**

Note: Teaching Assistants possess the authority to **reject a dataset and request a replacement**.

Quality control

- **Clean the Dataset:** Apply the appropriate cleaning process to ensure that your dataset is free from errors, inconsistencies, and irrelevant data.
- **Validation Schema:** Develop a validation schema that is suitable for your dataset. This schema will help ensure that the data meets the required standards and is valid for analysis.

Data Dashboard

- After cleaning the dataset, **design a dashboard** for your **cleaned dataset**, using suitable graphs, charts and tables. You can use an **Excel table pivot, Power BI**, etc.

Submission will be through the **classroom**, and you need to **submit the following**:

1. **Your Code** (contains Profiling, Cleaning and Validation code).
2. **A copy of the attached Profiling Report, in PDF format.**
3. **The generated Profiling Report in HTML format.**
4. **The Cleaned Dataset Dashboard – PDF format.**
5. **A documentation report** that documents your work and its distribution at this phase.

Notes:

- Late deduction within **3 days: 50% deduction** from the original grade.
- After 3 days, the grade will be dropped to **zero**.
- All submissions, including code, documentation, and presentations, must reflect your own work and understanding.
- The use of AI-generated content (e.g., ChatGPT, GitHub Copilot, etc.) is not allowed
- Any violation of this policy will result in a grade of zero for the entire project.

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Phase 2: Privacy and Security Implementation

Implement the necessary privacy policies and protection procedures to protect the confidentiality and integrity of the dataset. This is important to safeguard sensitive information and comply with relevant regulations.

❖ Data Security and Encryption Implementation:

- Apply data security techniques to protect your data **using Python libraries** such as **cryptography**, **hashlib**, and **PyCryptodome** as follows:

- Use **hashing techniques** to protect sensitive information (such as: passwords, phone numbers, transaction IDs, Important IDs, etc.)
- Build the **RSA Encryption and Decryption** algorithms from **scratch** to encrypt the textual data you have in your dataset, then decrypt the output ciphertext. Print the output as follows:

Plain Text	Encrypted Text	Decrypted Text
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❖ Access Control and Privacy Policies

- Implement Attribute-based access control to ensure only authorized users with the appropriate policies (leaving it to you to decide on the suitable policy for your project) can access or modify data.

❖ Using Python for Regulatory Compliance

- In your documentation, write a chapter about Regulatory Compliance:
 - What is the Regulatory Compliance
 - What is the GDPR
 - What is the CCPA
 - What is HIPAA
 - Make a comparison between the three (GDPR, CCPA, HIPAA)
- Demonstrate compliance with data protection regulations by utilizing Python libraries for:

- CCPA: [ccpa, privacy-python, datasketches]
- HIPAA: [Hippo, pyTenable, SecurityMonkey]

Submission: It will be through the **classroom**, and you need to **submit your whole code (Phase 1 & 2), your Documentation report** that documents your work and its distribution at this phase, **and a Final Presentation** that outlines your project workflow

Documentation and Discussion:

Your team is required to:

1. **Document Your Work:**

Provide clear documentation of all the steps taken across all phases of the project. This includes your methodology, tools used, decisions made, and any challenges faced.

2. **Prepare a Final Presentation:**

Create a presentation that outlines your project workflow, team contributions, and the overall structure of your solution. Highlight key features and outcomes.

3. **Code and Concept Review:**

You will be asked to explain and discuss your code, as well as the concepts and reasoning behind your implementation. Be prepared to justify your approach and demonstrate your understanding.

Bonus 1 (2 points):

- ❖ Write a Python code that performs an **attack on the RSA encryption algorithm** that should **attack your substitution-encrypted ciphertext** and decrypt it again to its original plain text.
- ❖ You need to apply this attack on your ciphered data text that resulted from the substitution encryption algorithm that you did in phase 2, then show us the output plain text that will result from this attack and compare it with the original plain text from your data and observe if they are the same or different.

Bonus 2 (1 point):

- ❖ **Perform the needed preprocessing** for your data to enter it as an input for a suitably **designed machine learning model** by you that can benefit from your data in the most useful way.

Notes:

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